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EXTENDED ABSTRACT

Coupling Machine Learning and Data Assimilation for Ocean Eddy Parameters' Prediction

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1. Introduction and objectives

Data assimilation, which combines observations and forecasts from a dynamic model to estimate the state of a system, is a major tool in the field of geosciences. In the case of specific geophysical structures (GS) such as cyclones or ocean eddies, conventional assimilation in a grid-based system tends to distort the physical properties of these GSs due to differences in position and shape between GSs from observations and models [Chen(2007)]. In this study we focus on mesoscale surface ocean eddies that we model as simplified like-Gaussian 2D objects defined by a set of parameters $\boldsymbol{x}$. Then, we propose to work on an assimilation scheme based solely on these eddy parameters. Their time propagation is determined by a machine learning model. This assimilation in a reduced space reduces computational costs and preserves the properties of the studied GSs.

2. Methods and data

Eddy detection and tracking algorithms, like AMEDA by [LeVu(2018)], can extract the eddy centroids, their maximal velocity, polarity, maximal anomaly variation and their contours. We perform data assimilation on relevant parameters, $\boldsymbol{x}$, of those structures in a reduced parametric space (see Figure 1), whose dimension n corresponds to the number of selected parameters, here $n = 11$, as stated in the following

$$\boldsymbol{x}(t) = \boldsymbol{M}(t-1)\boldsymbol{x}(t-1) + \boldsymbol{\eta}, \tag{1}$$

where $\boldsymbol{M}(t-1)$ is the model matrix from $t-1$ to t and $\boldsymbol{\eta} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q})$ is the model stochastic error. This updated state vector is related to the observation by:

$$\boldsymbol{y}(t) = \mathcal{H}(\boldsymbol{x}(t)) + \boldsymbol{\varepsilon}, \tag{2}$$

where $\boldsymbol{y}$ is a vector containing the parameters of the observed GS, $\mathcal{H}$ is the observation operator, and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R})$ is the stochastic observation error. We suppose here that the system is fully observed, hence $\mathcal{H} = \boldsymbol{I}_n$.

We use simple explainable machine learning models to define the daily time propagation model $\boldsymbol{M}$ of the eddy structure properties. In this study, the training and test datasets are the AMEDA outputs in the Arabian Sea over the 1993-2022 period. Two approaches of the least square regression model are

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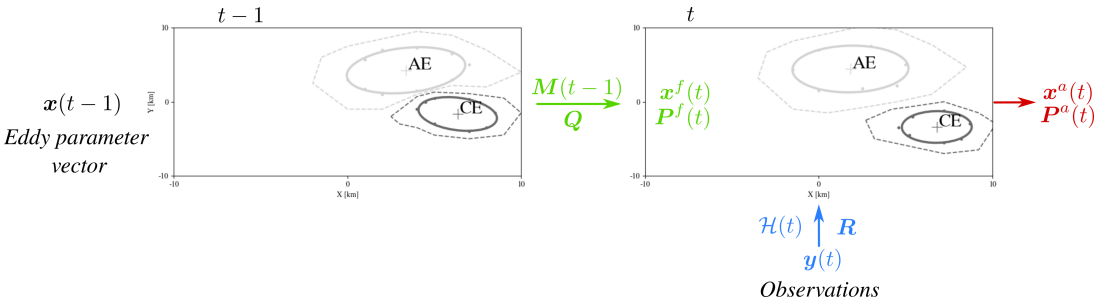

Figure 1. *Schematic diagram of the data assimilation of the parameters of a cyclonic (CE) eddy and an anticylonic (AE) eddy in the parametric space. The blue color indicates the observed information, the green one indicates the forecast information and the red one the analyzed information.*

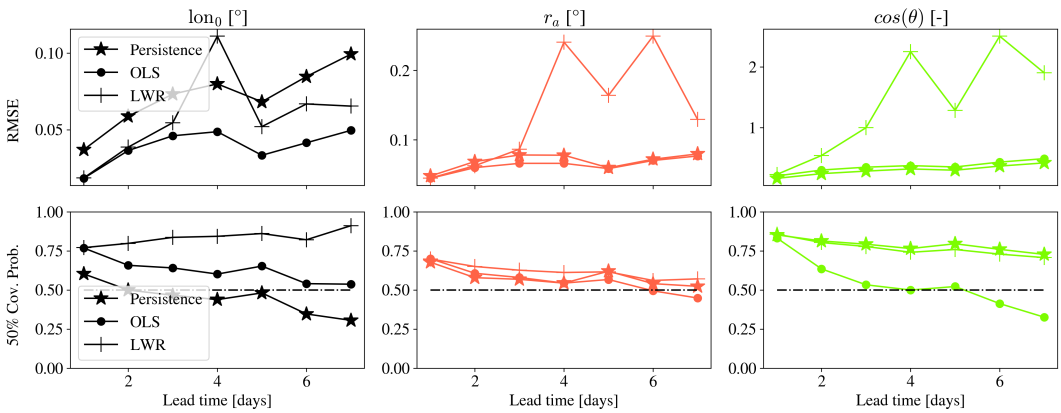

Figure 2. *Mean RMSE (top) and 50% Coverage Probability (bottom) of 52 eddies as a function of the lead time plotted for three eddy parameters: the centroid longitude x_0 [°] in black, the major semi-axis r_a [°] in red and the cosine of the rotation angle $\cos(\theta)$ [-] in green. The dots indicate the OLS model, the crosses the LWR model and the triangles the persistence baseline.*

trained: the global linear regression (OLS) on the whole training dataset and the local linear regression (LWR) based on analogs. Performances of each model are evaluated with the RMSE metric and the 50% coverage probability. The analysis step is performed with a linear Kalman filter and $(\boldsymbol{Q}, \boldsymbol{R})$ is optimized through the offline Expectation-Maximization algorithm.

3. Preliminary results and/or expected outcomes

The results show satisfying performances of both linear models over a one-week prediction with an assimilation rate of 5 days in Figure 2. Both models display similar performances, the OLS model having better performances in the long term and overcoming the persistence while the LWR model tends to diverge after 3 days. The interest of those models lies in the estimation of uncertainties and their propagation, which is integrated in a simple linear and Gaussian DA scheme.

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